ASSIGNMENT : 13

TITLE : Write a Java program to perform reading from and writing to a file using file handling operations.

Problem Statement : Write a Java program to perform reading from and writing to a file using file handling operations.

Learning Objectives : To perform file reading and writing operations using Java file handling classes.

Theory : Java provides file handling through the java.io package. Classes such as File, FileReader, FileWriter, BufferedReader, and BufferedWriter are used to perform input and output operations. File handling enables permanent storage and retrieval of application data. Proper exception handling is necessary to manage file-related errors. These concepts are essential for data persistence.

Exercise :

1. Create a program to write and **read employee details from a file.**

Welcome | java13.java 1

java13.java > Java > java13 > main(String[] args)

```java
import java.io.*;

public class java13 {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) throws IOException {

        FileWriter fw = new FileWriter(fileName: "employee.txt");
        fw.write(str: "Employee ID: 101\n");
        fw.write(str: "Name: Anushree Shinde\n");
        fw.write(str: "Department: IT\n");
        fw.write(str: "Salary: 50000\n");
        fw.close();

        System.out.println(x: "Employee details written successfully.\n");

        FileReader fr = new FileReader(fileName: "employee.txt");
        BufferedReader br = new BufferedReader(fr);

        String line;

        System.out.println(x: "Employee Details:");
        while ((line = br.readLine()) != null) {
            System.out.println(line);
        }

        br.close();
    }
}
```

OUTPUT :

```
Employee details written successfully.

Employee Details:
Employee ID: 101
Name: Anushree Shinde
Department: IT
Salary: 50000
```

java13.java 1 | employee.txt

employee.txt

```
1   Employee ID: 101
2   Name: Anushree Shinde
3   Department: IT
4   Salary: 50000
5
```

2. Create a Java program to write **patient details** (patient ID, name, age, and diagnosis) to a file and read the details from the file for display.

java13b.java > ...

```java
import java.io.*;

public class java13b {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) throws IOException {

        FileWriter fw = new FileWriter(fileName: "patient.txt");
        fw.write(str: "Patient ID: P101\n");
        fw.write(str: "Name: Rahul Sharma\n");
        fw.write(str: "Age: 35\n");
        fw.write(str: "Diagnosis: Fever\n");
        fw.close();

        System.out.println(x: "Patient details written successfully.\n");

        FileReader fr = new FileReader(fileName: "patient.txt");
        BufferedReader br = new BufferedReader(fr);

        String line;

        System.out.println(x: "Patient Details:");
        while ((line = br.readLine()) != null) {
            System.out.println(line);
        }

        br.close();
    }
}
```

OUTPUT :

```
Patient details written successfully.

Patient Details:
Patient ID: P101
Name: Rahul Sharma
Age: 35
Diagnosis: Fever
```

patient.txt

```
1   Patient ID: P101
2   Name: Rahul Sharma
3   Age: 35
4   Diagnosis: Fever
5
```

Outcome :

- Understood file handling operations in Java.
- Implemented programs to read from and write to files using Java I/O classes.
- Learned the importance of file handling for permanent data storage and retrieval.

Conclusion :

This assignment demonstrated file handling in Java using file input and output operations. Reading from and writing to files enables data persistence and efficient data management, making it an essential feature in real-world applications.